



Python Operators



1. Arithmetic Operators



Operator	Name	Example	Result
+	Addition	5 + 3	8
-	Subtraction	5 - 3	2
*	Multiplication	5 * 3	15
/	Division	5 / 2	2.5
//	Floor Division	5 // 2	2
%	Modulus	5 % 2	1
**	Exponentiation	5 ** 2	25

2. Comparison Operators



Operator	Name	Example	Result
==	Equal	5 == 5	True
!=	Not Equal	5 != 3	True
>	Greater Than	5 > 3	True
<	Less Than	5 < 3	False
>=	Greater or Equal	5 >= 5	True
<=	Less or Equal	5 <= 3	False

3. Assignment Operators

Operator	Equivalent To	Example	Result
=	x = value	x = 5	x = 5
+=	x = x + value	x += 3	x = x + 3
-=	x = x - value	x -= 3	x = x - 3
*=	x = x * value	x *= 3	x = x * 3
/=	x = x / value	x /= 3	x = x / 3
//=	x = x // value	x //= 3	x = x // 3
%=	x = x % value	x %= 3	x = x % 3
**=	x = x ** value	x **= 3	x = x ** 3

4. Logical Operators



Operator	Name	Example	Result
and	AND	(5 > 3) and (2 < 4)	True
or	OR	(5 > 3) or (2 > 4)	True
not	NOT	not (5 > 3)	False

5. Membership Operators



Operator	Name	Example	Result
in	In	'a' in 'apple'	True
not in	Not In	'z' not in 'apple'	True
in	In (List)	3 in [1, 2, 3, 4]	True
not in	Not In (List)	5 not in [1, 2, 3]	True

6. Identity Operators



Operator	Name	Example	Result
is	Is	a is b	True/False
is not	Is Not	a is not b	True/False

★ "is" checks if two variables reference the same object in memory, not if they are equal.

7. Bitwise Operators



Operator	Name	Example	Result
&	AND	5 & 3	1
	OR	5 3	7
^	XOR	5 ^ 3	6
~	NOT	~5	-6
<<	Left Shift	5 << 1	10
>>	Right Shift	5 >> 1	2

8. String Operators / Methods

Method	Example	Result	Description
+	'Hi ' + 'Python'	'Hi Python'	Concatenation
*	'ha' * 3	'hahaha'	Repetition
in	'Py' in 'Python'	True	Check if exists
not in	'py' not in 'Python'	False	Check if not exists
len()	len('Hello')	5	Length of string
.lower()	'PY'.lower()	'py'	Convert to lower case
.upper()	'py'.upper()	'PY'	Convert to upper case
.strip()	' hi '.strip()	'hi'	Remove spaces
.split()	'a,b,c'.split(',')	['a','b','c']	Split to list
.join()	'-'.join(['a','b','c'])	'a-b-c'	Join list to string
.replace()	'hi'.replace('i','y')	'hy'	Replace text
.find()	'hello'.find('e')	1	Find index
.count()	'banana'.count('a')	3	Count occurrence

Quick Examples

```
x = 10
x += 5      # x = 15
x *= 2      # x = 30
print(x > 20 and x < 40) # True
print('a' in 'apple')    # True
print(id(x) is id(x))    # True
```

Tips

Operators are symbols that perform operations on values and variables.

♥ Practice regularly, code daily, and enjoy Python!

